Handwritten Model and Explanation

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Research Question: How will the early expiration of Novo Nordisk's Canadian patent for semaglutide-based drug Ozempic affect the price and quantity of diabetes medication within Canadian brand and generic pharmaceutical markets?

Brief Market Definition

Product: Ozempic and its generic biosimilars

While I originally planned to study Novo Nordisk's 2 semaglutide-based drugs, Ozempic and Wegovy, no Canadian data was found for the latter. Ozempic on the other hand, while data is still scarce, has a few figures that can be drawn, such as cost per unit, price per unit, and total yearly sales. The large assumption being made is that there are no current substitutes for Ozempic. The focus of the model will be on the dynamic between Novo Nordisk and its generic competitors once they enter; hence, we will not touch upon other imperfect substitutes within the market. Any drugs prescribed for type-2 diabetes, such as Eli Lilly's Trulicity, that are not semaglutide based, will not be considered to have any effect in the model. In reality however, they would necessarily be included within the market since they impact the sales of Ozempic. The only competitors in the market will be bioequivalent generic drug producers, who sell a product identical in function and composition (semaglutide) to Novo Nordisk's Ozempic. None of these firms currently exist, but they are anticipated to enter the market within 2026.

Geography: Entire Canadian Market

Given the highly regulated market and the lack of any provincial data, we must define the geography at a national scale. Regulation on Canadian pharmaceuticals is conducted exclusively at the national level by the PMPRB; reports have shown that there are limited differences in drug prices between the various provinces.[1] Therefore, the quantity of Ozempic will be referring to that sold across the entire country. The Canadian market for pharmaceuticals is not directly impacted by the prices or demand in other countries, since each country is faced with differing legislature. The main international consideration would be the impact of imports prices and tariffs, since Novo Nordisk manufactures its drugs in both Denmark and the U.S. (North Carolina). While it is possible this could impact the cost curves over time, for simplicity we will not consider this effect. Controversies in British Columbia have arisen whereby American consumers were purchasing Ozempic from B.C. at a cheaper price than what was available to them domestically.[2] For simplicity in this model, the sole focus will be on Canadian consumers purchasing Ozempic within Canada.

Model Rationale and Explanation

I will model the market at three primary stages, all in relation to the entry of generics into the Canadian Ozempic market, which is slated to occur in 2026.

$\boldsymbol{t = -1}$: Before Novo Nordisk's Patent Expiration for Semaglutide – Monopoly (page 6)

Currently, it is assumed Novo Nordisk holds a monopoly in the semaglutide and Ozempic market. Since the patent has yet to expire, Novo Nordisk still has Government of Canada-granted exclusive rights to produce Ozempic. No other biosimilars are legally allowed to enter the market or be sold. This represents an unambiguous classic monopoly model, where there is one singular firm in the market and there is no entry possible. Novo Nordisk's monopolistic power is also evident from its ability to set price over 46 times higher than its marginal cost (see assumptions below) and the fact that it faces highly inelastic demand. In general, pharmaceutical products, especially those used to treat chronic illnesses such as type-2 diabetes, are considered essential goods. If Novo Nordisk raises its price, demand will fall, but by a smaller magnitude, providing Novo Nordisk its monopolistic power.

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[1] https://www.canada.ca/en/health-canada/programs/consultation-regulations-patented-medicine/document.html
[2] https://www.cbc.ca/news/canada/british-columbia/ozempic-canada-british-columbia-how-it-works-1.6794950

$\boldsymbol{t = 0}$: This will occur in 2026, when generic firms are projected to first enter into the market. No model will be created for this point in time.

$\boldsymbol{t = 1}$: Short-Term Behaviour of Novo Nordisk After Semaglutide Patent Expiration
– Generics Paradox (Application of Monopoly) (pages 7-8)

In the short run, there remains high brand loyalty, particularly for customers with chronic illnesses such as diabetes. From a behavioural perspective, consumers are unwilling to immediately switch from brand to generic. This often stems from consumers 1) believing the quality of the generic is inferior to that of the brand, 2) having asymmetric information, in which physicians still advertise and prescribe the brand-name drug, in addition to the brand firm having higher advertising capabilities, or 3) having taken the brand name drug for a significant amount of time that they do not see a point in alternating (which is often the case with the elderly population). This creates a significant price inelasticity of demand for brand Ozempic. Hence, even when the generic drugs have entered into the market at a cheaper price, many consumers will remain loyal to Ozempic. Those with more price elastic demand will shift to the generics, decreasing the total demand for brand name Ozempic, but also leaving only the most inelastic consumers. As described later, this represents a leftward shift in the demand curve, but also a steepening. Less elasticity will result in increased market power. Therefore, a monopoly model is still applicable in the short run, even once the generic firms have entered the market. This yields the generics paradox, commonly discussed in the literature (e.g. Vandoros & Kanavos, 2013), whereby the brand name is able to use its market power to raise prices even further once generic drugs have drawn away price elastic consumers.

$\boldsymbol{t = 2}$: Long-run Equilibrium After Patent Expiry
– Competitive Fringe Model with Dominant Firm (pages 8-10)

In practice, it would take many years to reach this equilibrium. Literature suggests that most adjustment to this state will be completed by eight years after the patent expiry, but it may be even longer than this (e.g. Duflos & Lichtenberg, 2011). By this time, brand loyalty has experienced a significant decline. Consumers are now aware of the presence of generics and equally aware that these generics are identical in composition to the original brand name drug. Hence, with the presence of perfect substitutes, the brand Ozempic price will fall downward in order to maintain competitiveness. An increasing number of consumers will transition towards generics at first when their price is cheaper. Nonetheless, empirical evidence suggests the brand drug maintains a sizeable market share, even long after the patent expiry (Ito, Hara & Kobayashi, 2020). We claim that Novo Nordisk is the dominant firm in the market. This is the result of their lowering prices to attract consumers, some enduring brand loyalty, and their large production capacity. While market power will be limited for Novo Nordisk, some degree will still persist in the future. In the long run, Novo Nordisk will need to take into consideration the amount being produced by the generics, in order to set its profit maximizing quantity, and the corresponding price. It no longer has all the demand, but rather the residual. Generic firms have little to no market power and are known to be price takers. Individually, each of their market shares will be only a fraction of the brand firm. They will produce in accordance with their costs and the price of the dominant brand. While not entirely realistic, for this model we will assume that the generic firms are perfectly competitive and that they will only take the price set by Novo Nordisk. Therefore, within our model the generic firms represent the competitive fringe.

Within this model there is also the extension to government regulations. The Government of Canada is known for putting price controls on generics in particular, assuring there is cheap access to medications for consumers. Under this assumption, if the price ceiling is binding, then the competitive fringe will take the price set by the government and produce accordingly, and Novo Nordisk will factor the resulting generic supply into its residual demand.

Assumptions

Price of Ozempic in Canada, 2024: $300 per month – which includes four doses

There is a large range of sources online claiming different prices Ozempic. This of course fluctuates slightly from province to province. CBC claims it is around $300,[3] others claim between $200-$300,[4] and another source saying up to $400.[5] Hence, $300 appears to be a consistent mean price point in Canada. We will make the assumption that this is the blanket price of Ozempic across Canada in 2024. The PMPRB does not release its regulation to the public, so we do not know what price controls have been set on Ozempic. This $300 may be assumed to be that price control, although we do not know this with certainty. The monopoly model for $t = -1$ will first be analyzed without a price control, but this can easily be added in afterwards.

Marginal Cost of Ozempic Production: $6.5 per month

It was reported that the cost to make Ozempic was around $4.73 USD per month in 2024.[6] Using the Bank of Canada's 2024 exchange rate 1.3698 CAD/USD,[7] we would say the cost is approximately $6.5 CAD. This of course is the average cost; however, since Novo Nordisk no longer has R&D costs for Ozempic, and it since it is impossible to determine what other costs are attributable to Ozempic production in Canada alone (since Novo Nordisk produces a plethora of other drugs), we will assume that MC=AC. This therefore relies on the large assumption that marginal cost will be constant for monthly Ozempic production. This helps to simplify the model, but is also plausible. For a large producer such as Novo Nordisk, and since the costs of active pharmaceutical ingredients, excipients, and packaging are all relatively cheap, the brand producer can easily scale production without incurring higher costs. Further assumption will be imposed that this marginal cost remains constant over time (in the $t = \{-1, 1, 2\}$ models). This is also plausible since Novo Nordisk has already achieved economies of scale with its production and it has well-established processes that will not fluctuate in cost.

Number of Monthly Units Sold in Canada: 694,444 units per month

It was reported that over $2.5 billion in sales were made in Canada in 2024.[8] If we divide this by the 12 months, we get find a monthly sale of around $208,333,333. We can then take the total monthly sales and divide again by the assumed price of $300 per month, yielding approximately 694,444 units of Ozempic sold per month. Many simplification assumptions are made here, such as equal sales across each month (where in reality they were likely growing) and that each customer pays the identical $300. Perhaps the most grandiose assumption of all is that it does not matter whether consumers are paying out of pocket, or whether it is paid by a public insurance company (which for example is the case for seniors above the age of 65 in Ontario, or a private insurance company. While unrealistic, for simplification of the model, there will only be one type of consumer, which combines the three cases mentioned above, each paying the same price and behaving identically (i.e. they have the same price elasticity).

Sales Growth (gauge for increasing demand): 10% per year

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[3] https://www.singlecare.com/blog/ozempic-canadian-pharmacy/
[4] https://www.getmaple.ca/blog/semaglutide-ozempic-cost-in-canada/
[5] https://www.ctvnews.ca/canada/article/ozempic-set-to-become-more-affordable-accessible-in-canada-raising-concerns-of-drug-abuse/
[6] https://www.cnbc.com/2024/03/27/novo-nordisk-ozempic-can-be-made-for-less-than-5-a-month-study.html
[7] https://www.bankofcanada.ca/rates/exchange/annual-average-exchange-rates/
[8] https://fortune.com/2025/06/17/novo-nordisk-ozempic-wegovy-semaglutide-canada-patent-protection-fee/

Growth is stimulated by increasing cases of type-2 diabetes and also heightened awareness of the drug treatment. Ozempic in particular has had a strong presence in media within the last couple years. In 2025, Canadian sales reached $1.6 billion by the end of July, and it is forecasted that Novo Nordisk will make nearly $3 billion in Canada over the whole year.[9] This marks approximately a ($3 billion - $2.5 billion)/($2.5 billion) = 20% yearly growth. This growth however would be unsustainable. Global estimates assume the Ozempic market will nearly double in size from 2025 to 2032 (22.25 billion USD to 43.53 billion USD), marking a CAGR of 10.1%.[10] Since our competitive fringe with dominant firm model takes place in the "long-run", for the number years into the future that is selected for analysis, we will expand demand by a simplified 10% yearly. That is, for the $t = 2$ model, the monthly demand will be $(1 + 0.1)^n$ times larger n years into the future.

Total possible diabetes-related demand for Ozempic in 2024: 2,595,420

In 2024, the total number of Canadians, reported to have been diagnosed with diabetes by a health professional was 2,883,800.[11] According to Health Canada: 90% of diabetes cases are type-2.[12] So, the total number of patients with diagnosed type-2 diabetes in Canada is approximately 0.9 x 2,883,880 = 2,595,420. This would be an upper bound for diabetes-related total demand, although we would anticipate much lower than this, since not everyone with type-2 diabetes will automatically be prescribed Ozempic or choose to take it. Nonetheless, there rising demand is also largely due to the weight-loss aspect of the drug, which has gained significant traction in recent years, and is difficult to quantify.

Marginal cost for generics

It is unlikely that generics will be able to produce the drug at the constant $6.5 average and marginal cost obtained by Novo Nordisk, who currently benefits from its economies of scale. We do however know with certainty that generics will have a lower fixed cost, since they never need to undergo the same intensive R&D to develop the new biosimilars, but rather can copy what as already been done. They also have much less regulatory costs if they can prove bioequivalence – the drug works in an identical way to the brand name and has the same composition. Nonetheless, contrary to our assumption of Novo Nordisk's constant marginal cost, we will assume that the generics will be faced with upward sloping marginal cost. Since they have not yet optimized production, an attempt to scale up their business and produce large values of output will result in rising costs. Ozempic biosimilars have yet to enter the market, hence only postulations can be made. As discussed later, an important part of this model will be scenario analyses, in which different values are tested. This will be the case for generic marginal costs, and their impacts on the competitive fringe market structure will be analyzed. As a base case used within this assignment, I have selected the generic marginal cost function $MC = 6.5 + 0.001q_f$, where q_f is the monthly output of one generic producer. This suggests that it is able to produce a low output at roughly the same marginal cost as Novo Nordisk, since the costs of input and packaging will be identical. However, for every 1,000 more units produced, the marginal cost will rise by a dollar, since it is costly for generics to expand their production beyond capacity.

A similar assumption will be made that all n generic firms in the market will have identical cost structures. That is, the marginal costs will not differ between any generic production. Since the inputs are the same across all firms, and their production processes will also be similar, this assumption is not entirely unrealistic.

[9] https://www.bnnbloomberg.ca/business/economics/2025/10/17/the-potential-is-phenomenal-drug-maker-vimmy-pharma-plans-to-produce-made-in-canada-generic-version-of-ozempic/
[10] https://www.coherentmarketinsights.com/industry-reports/ozempic-market
[11] https://www.statista.com/statistics/434090/number-of-canadians-reporting-being-diagnosed-with-diabetes/
[12] https://www.canada.ca/en/public-health/services/publications/diseases-conditions/framework-diabetes-canada-infographic.html

Linearity of Demand Function

This assumption is made for the overall simplicity and understanding of the three models. In reality, the demand for Ozempic is certainly not linear, and consumers would be more or less responsive to price changes depending on the current value of price. Modeling a curved demand function, and then a curved residual demand for the competitive fringe model, would perhaps take away from the core focus of this analysis and become highly complex. Hence, while unrealistic, all models will assume linear demand of the form $a - bQ$.

References

Duflos, Gautier, and Frank R. Lichtenberg (2011) "Does competition stimulate drug utilization? The impact of changes in market structure on US drug prices, marketing and utilization," *International Review of Law and Economics* 32(1), 95-109

Ito, Yuki, Konan Hara, and Yasuki Kobayashi (2020) "The effect of inertia on brand-name versus generic drug choices," *Journal of Economic Behavior & Organization* 172, 364-379

Nguyen, Nguyen Xuan , Steven H. Sheingold, Wafa Tarazi, and Arielle Bosworth (2022) "Effect of Competition on Generic Drug Prices," *Applied Health Economics and Health Policy* 20(2), 243 -253

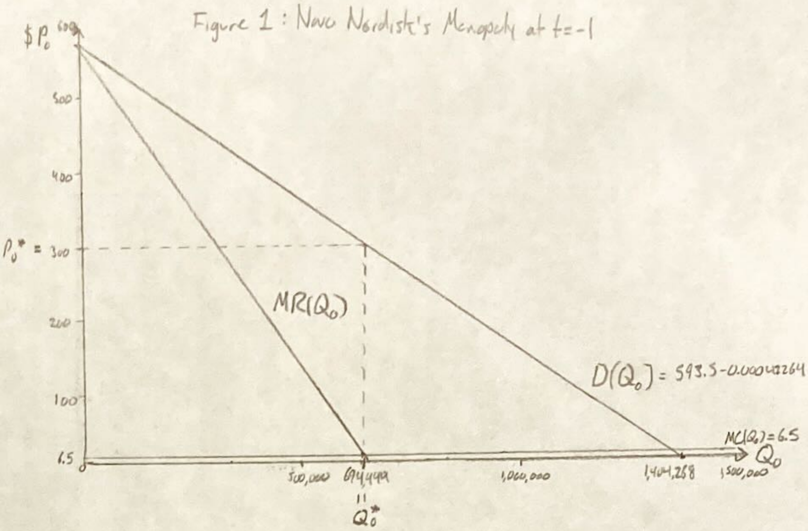
Vandoros, Sotiris, and Panos Kanavos (2013) "The generics paradox revisited: empirical evidence from regulated markets," *Applied Economics* 45(22), 3230–3239

$t=-1$: Before Novo Nordisk's Patent Expiration For Semaglutide – Monopoly

Variables and Functions:

- Q_0 = units of Ozempic per month in Canada } 1 monthly unit = 4 doses
- P_0 = price per unit of monthly Ozempic in CAD
- Q_0^* = profit maximizing quantity of Ozempic
- P_0^* = Ozempic price at profit maximizing quantity
- $D(P_0)$ = market demand for Ozempic in Canada
  - ↳ $D(Q_0)$ = inverse market demand for Ozempic in Canada
- $MC(Q_0)$ = marginal cost for Novo Nordisk's production of Ozempic
- $C(Q_0)$ = Novo Nordisk's monthly cost function for Ozempic
- $R(Q_0)$ = Novo Nordisk's monthly revenue function for Ozempic
- $MR(Q_0)$ = marginal revenue for Novo Nordisk's production of Ozempic
- $\pi(Q_0)$ = Novo Nordisk's monthly profit function for Ozempic

Figure 1: Novo Nordisk's Monopoly at $t=-1$



Profit Maximization

From Novo Nordisk's profit function, $\pi(Q_0) = R(Q_0) - C(Q_0)$, they select a profit maximizing quantity of monthly Ozempic to produce in Canada, Q_0^*, by $\frac{d\pi(Q_0^*)}{dQ_0} = \frac{dR(Q_0^*)}{dQ_0} - \frac{dC(Q_0^*)}{dQ_0} = 0 \Rightarrow MR(Q_0^*) = MC(Q_0^*)$ (i.e. the quantity where marginal revenue equals marginal cost). Note, the second order condition, $\frac{d^2\pi(Q_0^*)}{dQ_0^2} = \frac{d^2R(Q_0^*)}{dQ_0^2} - \frac{d^2C(Q_0^*)}{dQ_0^2} < 0$ must hold for profits to be maximized at Q_0^*.

The price of Ozempic at this profit maximizing quantity is found by $P_0^* = D(Q_0^*)$.

Here we anticipate the largest profits obtained by Novo Nordisk.

Estimations for Novo Nordisk's Ozempic Monopoly in Canada

- As stated above, we assume a linear market demand for Ozempic, of the inverse form $D(Q_0) = a - bQ_0$.
- From the limited data present, we found that $Q_0^* = 694{,}444$, $P_0^* = \$300$, and $MC(Q_0) = 6.5$ is a constant marginal cost. Hence, we already know the profit maximizing number of monthly Ozempic units, but we can use this to provide a rough estimate for the 2024 market demand function (which will be helpful for models at $t=1$ and $t=2$).

Novo Nordisk's revenue can be calculated as $R(Q_0) = P_0 \cdot Q_0 = D(Q_0) \cdot Q_0 = (a - bQ_0)Q_0 = aQ_0 - bQ_0^2$. Thus $R(Q_0) = aQ_0 - bQ_0^2$. Taking the derivative with respect to Q_0 we can calculate marginal revenue for Novo Nordisk as $MR(Q_0) = \frac{dR(Q_0)}{dQ_0} = a - 2bQ_0$.

How do we solve for a and b?

We know one point on the demand curve (694,444, 300) and one on the marginal revenue curve when it intersects marginal cost, (694,444, 6.5). We have two equations – $R(Q_0)$ and $MR(Q_0)$ – and two unknowns – a and b.

① When $Q_0 = 694{,}444$, $MR(Q_0) = 6.5 \Rightarrow MR(Q_0) = a - 2bQ_0 \Rightarrow 6.5 = a - 2b(694{,}444) \Rightarrow a = 6.5 + 2b(694{,}444)$

② When $Q_0 = 694{,}444$, $D(Q_0) = 300 \Rightarrow D_0(Q_0) = a - bQ_0 \Rightarrow 300 = a - b(694{,}444)$

Solving the system by substituting ① into ② we get $300 = (6.5 + 2b(694{,}444)) - b(694{,}444) \Rightarrow 300 = 6.5 + 694{,}444b \Rightarrow b = 0.00042264$

and hence $a = 6.5 + 2(0.00042264)(694{,}444) = 593.5$.

$\therefore a = 593.5$ and $b = 0.00042264$ so that $D(Q_0) = 593.5 - 0.00042264Q_0$.

Note: as a check for plausibility, if we take $P_0 = 0$, we find that $0 = 593.5 - 0.00042264Q_0 \Rightarrow Q_0 = 1{,}404{,}268$. This implies that if Ozempic were free, 1,404,268 Canadians would demand it each month. We found above that approximately 2,595,420 adults in Canada have diagnosed type-2 diabetes, many of whom take other medications or do not want to be on Ozempic, so our 1,404,268 figure is entirely plausible.

We could also rearrange this to the form $D(P_0) = \frac{593.5 - P_0}{0.00042264} \approx 1404268.4 - 2366.08P_0$, which implies that an increase of \$1 in the price of Ozempic will roughly decrease demand by 2,366 units.

$t=1$: Short-Term After Ozempic Patent Expiration - Generic Paradox (Monopoly)

Variables and Functions:

- All variables and functions from $t=-1$ will carry over
- α = proportion of market share held by generics $0 \le \alpha \le 1$, although $\alpha \ge 0.75$ is unlikely
- $D_2(Q_o)$ = the new market demand for <u>brand Ozempic</u> only, following generic entry (inverse of $D_2(P_o)$).
- $R_2(Q_o)$ = the new monthly revenue function for Novo Nordisk following generic entry
- $MR_2(Q_o)$ = new marginal revenue for Novo Nordisk's production of Ozempic following generic entry
- $Q_o^{*'}$ = profit maximizing quantity of Ozempic following generic entry (short run)
- $P_o^{*'}$ = Ozempic price at profit maximizing quantity following generic entry
- Note: all costs are assumed to stay the same

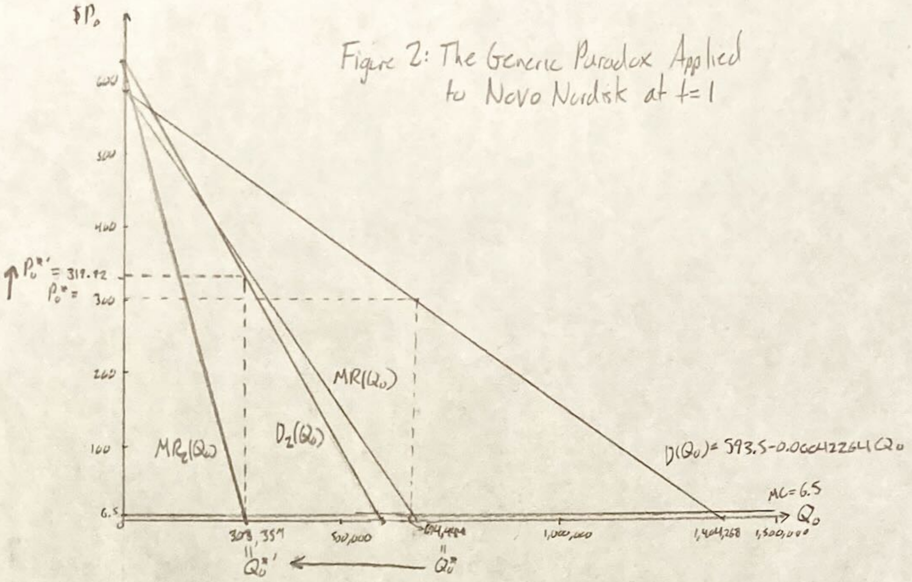


Figure 2: The Generic Paradox Applied to Novo Nordisk at $t=1$

New Market Demand for Brand Ozempic: Assume a new demand function of the form $D_2(Q_o) = a' - b'Q_o$

Since α share of the market belongs to generics, they will take the customers who are most elastic to price (since generics prices will be much lower upon entry). Therefore, the brand Ozempic will only have left its most loyal customers, those who are willing to pay higher prices in order to keep using the brand. This signifies that the remaining market demand for brand Ozempic will be more <u>inelastic</u> (graphically represented by a steep demand curve).

We have two movements in the demand for Ozempic: ① shift left - decrease in demand due to α share of market held by generics. Ozempic now held only $(1-\alpha)$ of the market.
② rotation to be steeper - remaining demand is more inelastic

Profit Maximization under new brand demand for Ozempic

We know from the textbook and lecture videos, that when a firm is faced with inelasticity or low elasticity of demand, they are able to achieve market power. This observed through equations such as $\frac{P}{MC} = \frac{1}{1+\frac{1}{\varepsilon}}$ or the Lerner Index $\frac{P-MC}{P} = -\frac{1}{\varepsilon}$. Hence, under the new market demand for brand Ozempic, Novo Nordisk will still maintain market power. Essentially, while it still has high brand loyalty (which we assume in the short run), Novo Nordisk operates as a monopoly, just facing a different demand curve, once some elastic consumers have already transitioned to generics.

Profit maximization strategy remains the same as in $t=-1$: $\pi_2(Q_o) = R_2(Q_o) - C(Q_o)$ is the profit of Novo Nordisk (with identical costs to before) and they select a profit maximizing quantity of monthly Ozempic to produce in Canada, $Q_o^{*'}$, by $\frac{d\pi_2(Q_o^{*'})}{dQ_o} = \frac{dR_2(Q_o^{*'})}{dQ_o} - \frac{dC(Q_o^{*'})}{dQ_o} = 0$ $\Rightarrow MR_2(Q_o^{*'}) = MC(Q_o^{*'})$ (i.e. the quantity where marginal revenue equals marginal cost). Again, the second order condition must also hold for profits to be maximized at $Q_o^{*'}$: $\frac{d^2\pi_2(Q_o^{*'})}{dQ_o^2} = \frac{d^2R(Q_o^{*'})}{dQ_o^2} - \frac{d^2C(Q_o^{*'})}{dQ_o^2} < 0$. The new price of Ozempic in the short run is found by $P_o^{*'} = D_2(Q_o^{*'})$

Empirical Estimates for the New Brand Ozempic Demand Curve.

First note, at $t=-1$, the elasticity at the profit maximizing quantity and price was: $\varepsilon_1 = \frac{dQ_o}{dP_o}\frac{P_o^*}{Q_o^*} = \frac{-1}{b}\frac{P_o^*}{Q_o^*} = \frac{-1}{0.00042264}\frac{(300)}{(694{,}444)} = -1.022$

↳ this could equally be obtained from $\frac{P}{MC} = \frac{1}{1+\frac{1}{\varepsilon}} \Rightarrow \frac{300}{6.5} = \frac{1}{1+\frac{1}{\varepsilon}} \Rightarrow \varepsilon = -1.022$ as before.

For our new demand curve, $D_2(Q_o) = a' - b'Q_o$, we know that at $P_o^* = 300$, $|\varepsilon|$ must be smaller in magnitude than 1.022.

Hence, we may estimate b' by $\frac{-1}{\varepsilon_2}\frac{P_o^*}{(1-\alpha)Q_o^*}$ (rearranging the formula from above).

↳ $(1-\alpha)Q_o^*$ represents the market share held by the brand name drug after generic entry. ★ We may also input $(1-\alpha)Q_o^*(1+g)$ where g is the growth in total demand following entry of generics. For example, $g=0.1$ to represent the estimated 10% yearly growth in Ozempic demand.

↳ ε_2 is the new assumed elasticity of demand for Ozempic brand drug following generic entry, $|\varepsilon_2| < 1.022$. Within my final paper, I will perform a senario analysis over different values of ε_2, but for now I will assume $\underline{\underline{\varepsilon_2 = -0.9}}$ since literature suggests that brand price elasticity may range from -0.3 to -1. (e.g. Vandoros and Kanavos, 2013)

↳ Assume $\alpha = 0.54 \Rightarrow 1-\alpha = 0.46$. Literature suggests that 2-4 years after entry, generics may compose around 54% of the market share. Assume $g=0$ for now, for simplicity. Again, this is something I will test in a sensitivity analysis. (e.g. Duflos and Lichtenberg, 2011)

For now, we can estimate $b' = \frac{-1}{-0.9}\frac{300}{(1-0.54)(694{,}444)} = 0.001043479$.

Since we know the point $([(1-0.54)(694{,}444)], 300)$ on $D_2(Q_o)$ we can solve for a'.

$$D_2(Q_o) = P_o = a' - b'Q_o \Rightarrow 300 = a' - 0.001043479[(1-0.54)694444] \Rightarrow a' = 633.33$$

Therefore, we can estimate a hypothetical market demand for brand Ozempic, following generic entry, as

$$D_2(Q_o) = 633.33 - 0.001043479 Q_o.$$

This new market demand for brand Ozempic is plausible since b' is steeper than b, and it has been shifted leftward from $D(Q_o)$.
Under this postulated market demand for brand Ozempic, we can find the new profit maximizing condition

- $R_2(Q_o) = P_o \cdot Q_o = D_2(Q_o) \cdot Q_o = (633.33 - 0.001043479 Q_o) Q_o = 633.33 Q_o - 0.001043479 Q_o^2$
- Taking the derivative to get marginal revenue yields $MR_2(Q_o) = \frac{dR_2(Q_o)}{dQ_o} = 633.33 - 0.002086958 Q_o$
- At the profit maximizing quantity $MR_2(Q_o^{*\prime}) = MC(Q_o^{*\prime}) \Rightarrow 633.33 - 0.002086958 Q_o^{*\prime} = 6.5 \Rightarrow Q_o^{*\prime} \cong 300{,}357$
- The corresponding price in $P_o^{*\prime} = D_2(Q_o^{*\prime})$ is $P_o^{*\prime} = 633.33 - 0.001043479(300{,}357) = \$319.42$

Therefore, under the new demand for brand Ozempic, after generic entry, the quantity of brand Ozempic may be 300,357 per month, sold at a price of \$319.42.

↳ Again, these figures will be explored in sensitivity analysis for different values of ε_2, α, and g. (e.g. Vandoros, Kanavos, 2013)

- The new price of \$319.42, a 6.64% increase from before generic entry, is perfectly in line with the roughly 5% figures noted in the literature, and hence some degree of generic paradox can be displayed using this model.
  As expected, the quantity of Ozempic, $Q_o^{*\prime} = 300{,}357$, falls drastically as the generic share, α, increases, but also as Novo Nordisk raises its prices.

$t=2$: Long-run Equilibrium After Ozempic Patent Expiry
Competitive Fringe Model with Dominant Firm

Now we assume we are in the long run, where brand loyalty has declined so that Ozempic, while still dominant, no longer has the same degree of price elasticity. The generics will act as price takers since there will be rising competitiveness between them. Novo Nordisk will take this supply into account while deciding its optimal output, and market price will also be set accordingly.

Variables and Functions

- q_{fi} = units of monthly Ozempic biosimilar produced by generic firm i, $1 \le i \le n$
- $MC_i(q_{fi})$ = marginal cost for generic firm i to produce Ozempic biosimilar
- Q_f = total monthly Ozempic biosimilars produced by generic firms; $Q_f = \sum_{i=1}^{n} q_{fi}$ i.e. it is the sum of each generic firm's output
- P_o = market price of Ozempic and its biosimilars
- $s_i(P_o)$ = supply function of monthly Ozempic biosimilar by generic firm i, $1 \le i \le n$
- $S_f(P_o)$ = supply function of monthly Ozempic biosimilars by all generic firms
  - ↳ if each generic firm is identical, we can take $S_f(P_o) = n \cdot s_i(P_o)$ ← we have made this assumption above
  - ↳ if each generic firm is different (i.e. heterogenous marginal costs) then $S_f(P_o) = \sum_{i=1}^{n} s_i(P_o)$
- Q_{NN} = units of monthly brand Ozempic produced by Novo Nordisk
- Q_o = total units of monthly brand and biosimilar Ozempic produced in the market; $Q_o = Q_f + Q_{NN}$
- $D(P_o)$ = market demand for brand and biosimilar Ozempic
- $D_{NN}(P_o)$ = market demand for brand Ozempic only; the difference between total market demand and total generic supply, that is $D_{NN}(P_o) = D(P_o) - S_f(P_o)$
- $MC_{NN}(Q_{NN})$ = marginal cost for Novo Nordisk's production of Ozempic
- $R_{NN}(Q_{NN})$ = Novo Nordisk's monthly revenue function for Ozempic
- $MR_{NN}(Q_{NN})$ = marginal revenue for Novo Nordisk's production of Ozempic
- Q_{NN}^* = profit maximizing quantity of brand Ozempic for Novo Nordisk given the residual demand
- P_o^* = price corresponding to Novo Nordisk's Q_{NN}^*, i.e. $D_{NN}(Q_{NN}^*)$ – will be the price generics take as well.

How the model works:

① Competitive Fringe:

- Each of the n generic drug producers in the market will be price takers, so they will each choose output q_{fi} such that $P_o = MC_i(q_{fi})$. Marginal cost will of course be derived from each generic firm's total cost function, $MC_i(q_{fi}) = \frac{dC_i(q_{fi})}{dq_{fi}}$
- Rearranging this, we will find $q_{fi} = s_i(P_o)$, the supply for each individual generic producer.
- Lastly, total supply of Ozempic biosimilars will be the horizontal sum of each of the individual supply curves:
  $S_f(P_o) = \sum_{i=1}^{n} s_i(P_o) = n \cdot s_i(P_o)$ (since we are assuming each generic firm is faced with the same costs).

② Novo Nordisk sets quantity and price based on the total generic firm supply curve:

- Novo Nordisk no longer has access to the entirety of demand, but rather the residual demand after generic drugs have been supplied. Therefore, $D_{NN}(P_o) = D(P_o) - S_f(P_o)$; that is, the residual demand for brand Ozempic is the total market demand minus the total generic supply.
- To calculate Novo Nordisk's marginal revenue, $MR_{NN}(Q_{NN})$, the inverse residual demand for brand Ozempic must be calculated. Hence, $Q_{NN} = D_{NN}(P_o) \Rightarrow P_o = D_{NN}(Q_{NN})$.
- Now, Novo Nordisk's revenue function from monthly Ozempic in Canada is $R_{NN}(Q_{NN}) = P_o \cdot Q_{NN} = D_{NN}(Q_{NN}) \cdot Q_{NN}$, and by consequence its marginal revenue will be $MR_{NN}(Q_{NN}) = \frac{dR_{NN}(Q_{NN})}{dQ_{NN}}$
- Parallel to both $t=-1$ and $t=1$, Novo Nordisk maximizes its profit, $\pi_{NN}(Q_{NN}) = R_{NN}(Q_{NN}) - MC_{NN}(Q_{NN})$, by selecting Q^*_{NN} in accordance to the first order condition $\frac{d\pi_{NN}(Q^*_{NN})}{dQ_{NN}} = \frac{dR(Q^*_{NN})}{dQ_{NN}} - \frac{dC(Q^*_{NN})}{dQ_{NN}} = 0 \Rightarrow MR_{NN}(Q^*_{NN}) = MC_{NN}(Q^*_{NN})$. Once again the second order condition must hold as well.
- Finally, the market price for Ozempic is determined by inputting Q^*_{NN} into the residual demand for brand Ozempic. i.e. $P^*_o = D_{NN}(Q^*_{NN})$.

③ Competitive Fringe takes this value as given to determine output.

- Total generic drug supply is given by $S_f(P^*_o)$, at price P^*_o. We have $S_f(P^*_o) = Q^*_f$.
- To determine the supply of generics by each individual generic firm, assuming homogenous costs, we have $q_{fi} = \frac{S_f(P^*_o)}{n}$.

Note, let $\bar{P}$ be the price such that $S_f(\bar{P}) = 0$ (the price at and below which generic producers will not supply any Ozempic bioequivalents). If $P^*_o \leq \bar{P}$, then there will be no generics in the market and the brand Ozempic will be the sole producers with $D(P_o) = D_{NN}(P_o)$. This would occur if Novo Nordisk's marginal cost was significantly lower than any generics. However, we assumed above that this would not be the case, so for the most part we will observe $\bar{P} < P^*_o$.

IMPORTANT: graphs below display the problem in general, not with the specific numbers. Showing a $MC_{NN}(Q_{NN}) = 6.5$, in addition to the large total output of Ozempic became much too messy to read. The logic behind the calculations on the following page mirrors the graphs exactly, but is just at a much different scale. That is to say, the graphs below are NOT to scale.

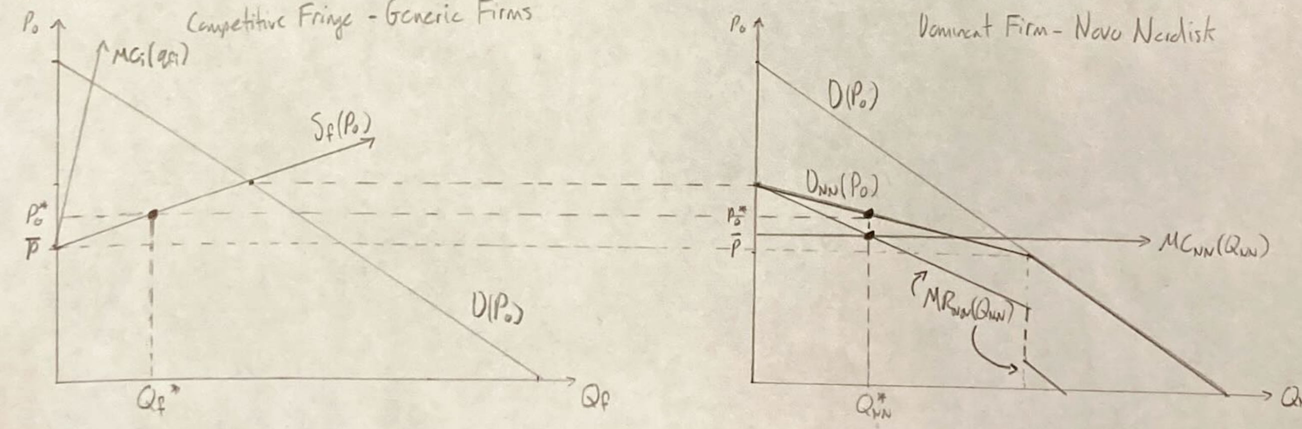


Figure 3: Competitive Fringe and Dominant Firm Model Applied to Generic Firms and Novo Nordisk

Empirical Application of Competitive Fringe Model with Dominant Firm to Ozempic in Canada in the long-run.

- From the assumptions, we know that Ozempic demand is expected to grow at around 10% each year until around 2032. Hence, we can assume this will result in a right-shift in the demand curve by a factor of $(1+g)^8 = (1+0.1)^8 = 1.1^8$. Here we are assuming that Canada will reach the "long-term" state in around 8 years. This is consistent with literature suggesting that generic market share settles around 8 years after patent expiry. Note, the 8 years is from 2024, when data is from! (eg. Wiles 2011; Nguyen et al., 2022)
- The other major assumption we need to make is the marginal cost (or total cost) of the generic firms. This must be purely speculative, since there is no real-world data yet for these firms entering the Canadian market. As above, I will eventually conduct a scenario analysis here, changing variables such as the generic marginal cost curve and the number of generic firms, n. For now, in accordance to introductory assumptions, so that there are diminishing marginal returns, we will set $MC_i(q_{fi}) = 6.5 + 0.001q_{fi}$ for all $i \in \{1,2,\dots,n\}$. Note, we assume $n=8$, in accordance to many generic drug markets from literature. (eg. Wiles et al 2011; Nguyen et al 2022)

Using this information, and the information derived in t=-1, we can solve for hypothetical values for Q_{NN}^*, Q_f^*, and P_o^*.

① i) $P_o = MC_i(q_{fi}) \ \forall i \in \{1,2,\dots,n\} \Rightarrow P_o = 6.5 + 0.001q_{fi} \Rightarrow q_{fi} = S_i(P_o) = \frac{P_o - 6.5}{0.001}$

ii) $S_f(P_o) = n \cdot S_i(P_o) = 8\left(\frac{P_o - 6.5}{0.001}\right)$

② i) $D(Q_o) = 593.5(1.1)^8 - 0.00042264Q_o$ since we are assuming an annual growth of 10% over 8-years

$\Rightarrow D(P_o) = \frac{593.5(1.1)^8 - P_o}{0.00042264}$

ii) $D_{NN}(P_o) = D(P_o) - S_f(P_o) = \frac{593.5(1.1)^8 - P_o}{0.00042264} - 8\left(\frac{P_o - 6.5}{0.001}\right)$ where $D_{NN}(P_o) = Q_{NN}$, so

$\Rightarrow 0.001Q_{NN} = 2.36608[593.5(1.1)^8 - P_o] - 8P_o + 52$

$\Rightarrow 0.001Q_{NN} - 3062.174046 = -10.36608P_o$

$\Rightarrow P_o = D_{NN}(Q_{NN}) = 295.4032813 - 0.00009646848\,Q_{NN}$ <- the residual demand for brand Ozempic

iii) $R_{NN}(Q_{NN}) = P_o \cdot Q_{NN} = 295.4032813Q_{NN} - 0.00009646848Q_{NN}^2$

$MR_{NN}(Q_{NN}) = \frac{dR_{NN}(Q_{NN})}{dQ_{NN}} = 295.4032813 - 0.000192937Q_{NN}$

iv) $MR_{NN}(Q_{NN}^*) = MC_{NN}(Q_{NN}^*) \Rightarrow 295.4032813 - 0.000192937Q_{NN}^* = 6.5$ (assuming Novo Nordisk maintains same marginal cost)

$\Rightarrow Q_{NN}^* = 1{,}497{,}397$ monthly units of Ozempic

v) $P_o^* = 295.4032813 - 0.00009646848(1497397.263) = \$150.95$

③ $Q_f^* = S_f(P_o^*) = 8\left(\frac{\$150.95 - 6.5}{0.001}\right) = 1{,}155{,}613$ monthly units of Ozempic biosimilar by total generics (or approximately 144,451 units of Ozempic biosimilar from each of the 8 generic firms).

Therefore, under these assumptions for generic marginal cost, demand growth in years to come, and the number of generics in the market, we can estimate that Novo Nordisk will produce around 1,497,397 monthly units of Ozempic and generics will in total provide 1,155,613 monthly units of a biosimilar. Market price would be \$150.95. This prediction is plausible, with a drastic fall in price from the current \$300, a large generic market share (43.5%), and increased production to meet larger demand.

☆ Addition of a Government of Canada Generic Price Ceiling ☆

This is a natural and simple extension to the Competative Fringe Model. In this case, if the price ceiling P_c is binding (i.e. $P_c < P_o^*$), generics will produce at the intersection between this ceiling and $S_f(P_o)$. i.e. $Q_f^* = S_f(P_c)$.

- The dominant firm, Novo Nordisk, will be subject to residual demand $D_{NN}(P_o) = D(P_o) - S_f(P_c)$, where $S_f(P_c)$ is known. Then as before, Novo Nordisk will maximize its profits based on $D_{NN}(P_o)$, finding Q_{NN}^* and P_o^*.

In Summary, if the price control on the generics is binding, they will produce $Q_f^* = S_f(P_c)$ at the price set by the government P_c, while Novo Nordisk will proceed as before with whatever residual demand is left. Their price, P_o^* will be above that of the price ceiling on the generics.

- Can be added to the model in the final report as a comparison.